

Standardized Faculty Grading Rubric & Assessment Matrix AY 2025-2026

School of Computing & Engineering | OBE Assessment, Lab Evaluation & Capstone Criteria

1. Institutional Assessment Standards & OBE Pedagogical Framework

The School of Computing and Engineering enforces an Outcome-Based Education (OBE) assessment model meeting National Board of Accreditation (NBA) Tier-1 criteria and aligned with the Washington Accord's graduate attribute taxonomy. OBE shifts the educational paradigm from input-based delivery to demonstrable learning outcomes: every assessment must explicitly measure defined Course Outcomes (COs) mapped to Programme Outcomes (POs) and Programme Specific Outcomes (PSOs). Bloom's Revised Taxonomy (2001) provides the cognitive depth hierarchy — Remember (L1), Understand (L2), Apply (L3), Analyze (L4), Evaluate (L5), Create (L6) — and each question in every assessment must be tagged with the appropriate Bloom's level and corresponding CO. The attainment of each CO is quantitatively measured through direct assessment (examination marks) and indirect assessment (exit surveys, peer evaluation) to drive continuous quality improvement mandated by the NAAC IDP.

Assessment at Northgate Institute is distributed across Continuous Internal Evaluation (CIE — 50% of total grade) and End-Semester Examinations (ESE — 50% of total grade). Course instructors must prepare question papers with explicit CO-PO-Bloom's mapping in the Question Paper Blueprint submitted to the Controller of Examinations 14 days before each examination.

2. Continuous Internal Assessment (CIE) Component Weights

For a standard 4-credit Professional Core Course (contact hours 3L-1T-2P per week):

- **CAT-1 (25% of CIE):** 90-minute centralized written exam covering Units I & II. Bloom's L2–L4. Q-paper: 5 questions × 6 marks = 30 marks scaled to 25. Faculty must upload marks to DCMP within 7 days of the test date.
- **CAT-2 (25% of CIE):** 90-minute centralized exam covering Units III & IV. Same structure and upload timeline as CAT-1.
- **Online Quizzes & Surprise Tests (15% of CIE):** Three auto-graded objective quizzes administered via DCMP LMS, each 20 questions (MCQ + Fill-in), covering the preceding 2 weeks of lectures. Average of best 2 out of 3 quizzes.
- **Homework Problem Sets & Analytical Proofs (15% of CIE):** Four individual written assignments requiring formal proofs, algorithm derivations, or design documents. Each assignment involves a mandatory plagiarism check (MOSS/Turnitin). Late submissions: 20% deduction per working day, no credit after 5 days.
- **Tutorial Participation & Board Work (10% of CIE):** Engagement in weekly problem-solving tutorials. Teaching assistants evaluate on a 3-point scale (Active Engagement, Partial Engagement, Absent). Attendance at 75% of tutorials is mandatory for tutorial credit.
- **Attendance Incentive (10% of CIE):** Pro-rated allocation: ≥ 95% = 10 pts; 90–94% = 8 pts; 85–89% = 6 pts; 80–84% = 4 pts; 75–79% = 2 pts; < 75% = 0 pts (and ineligible for ESE).

3. Programming Assignment & Laboratory Evaluation Matrix

Programming assignments (theory courses with lab components) are evaluated on five dimensions:

- **Functional Correctness (35%):** Code must pass all published test cases AND 8 hidden edge-case tests without runtime errors, incorrect output, or memory leaks. Full marks only if all tests pass; partial credit scaled to pass rate.
- **Algorithmic Complexity & Efficiency (20%):** Implementation must meet the optimal or near-optimal time and space complexity specified in the assignment brief. Asymptotic analysis (Big-O, Big-Theta, Big-Omega) with proof must be submitted in the README.
- **Software Design & Code Quality (15%):** Adherence to SOLID principles, clean modular decomposition, meaningful identifier naming (PEP8/Google style for Python; Google C++ Style for C++), appropriate comments, and absence of global state abuse or anti-patterns.
- **Documentation & Test Coverage (15%):** Comprehensive docstrings in Sphinx/Doxygen format, API contract documentation, and a unit test suite using pytest/Google Test achieving ≥ 85% branch coverage verified by coverage.py.
- **Academic Integrity (15%):** MOSS code similarity < 15% across all submitted solutions in the cohort. Submissions exceeding the threshold trigger DAIC review with potential zero-grade and further disciplinary action.

4. Laboratory Practical Examination & Comprehensive Viva Voce Rubric

End-semester laboratory examinations are jointly conducted by the Course Instructor and an External Examiner appointed by the Dean. A 3-hour practical session evaluates four components:

- **Pre-Lab Design & Documentation (20%):** Flowchart or pseudocode (Sem I–IV), UML class diagram or schema (Sem V–VIII), algorithm analysis, submitted in the lab record book 15 minutes before the session begins.
- **Implementation & Execution (40%):** Code compiles without errors, runs on the examiner-provided input, produces correct output for all test cases, and handles edge cases (empty input, maximum bounds).
- **Live Code Walkthrough & Debugging (20%):** Student explains their design, traces execution for a chosen input, and makes a minor modification requested by the examiner within 10 minutes.
- **Viva Voce Technical Defense (20%):** 5 oral questions covering underlying theory, design trade-offs, algorithm complexity, alternative approaches, and real-world applications. Depth and correctness of responses evaluated.

5. Mini-Project Evaluation Matrix (CS390, Semester V — 2 Credits)

Semester V teams of 2–3 students design, implement, and demonstrate a functional software system or embedded hardware-software prototype. Evaluation across 4 review milestones:

- **Review 1 — Problem Formulation & Feasibility (15%):** Problem statement clarity, domain literature review (≥ 5 academic/industry references), technology stack justification, project timeline (Gantt chart).
- **Review 2 — Architecture & Design (25%):** System architecture diagram, database ER schema or NoSQL data model, API contract (OpenAPI 3.0 spec), unit test plan.
- **Review 3 — Working Prototype Demonstration (40%):** Live demo of core features, code quality review, CI/CD pipeline demonstration, security assessment (basic OWASP scan).
- **Review 4 — Final Report & Presentation (20%):** IEEE-format technical report (≥ 8 pages), PowerPoint presentation, team contribution matrix, reflection on lessons learned.

6. Senior Capstone Project Evaluation (CS490, Semesters VII-VIII — 8 Credits)

B.Tech Capstone Projects are evaluated over two terms by a 3-member Faculty Review Panel (Primary Supervisor, Second Reader, External Examiner from industry or academia). Evaluation criteria:

- **Problem Formulation & Survey (15%):** Research problem novelty, comprehensive literature survey (≥ 20 references), identification of research gaps, measurable success criteria.
- **System Architecture & Design Spec (25%):** HLD (architectural diagram, service boundaries, data flow), LLD (class diagrams, sequence diagrams, DB schema), rationale for design decisions.
- **Engineering Implementation (30%):** Production-ready code with CI/CD pipeline, automated test suite ($\geq 90\%$ coverage), performance benchmarks, deployment documentation.
- **Technical Dissertation (15%):** IEEE-format thesis (≥ 40 pages), abstract, methodology, results, critical discussion, future work, Similarity Index $< 15\%$.
- **Oral Defense & Q&A; (15%):** 20-minute presentation + 15 minutes of panel Q&A; assessing depth of technical understanding and ability to defend design choices.

7. M.Tech Thesis & Research Seminar Assessment Rubric

M.Tech theses (20 credits) are evaluated by an External Examiner and the Thesis Supervisor. Key criteria: (a) Originality and contribution to knowledge; (b) Research methodology rigor; (c) Quality of experimental design and validation; (d) Clarity of technical writing; (e) Viva voce defense quality. Pre-submission similarity index must be $< 10\%$ (excluding references). The Annual Research Seminar (4 credits) is evaluated on: literature survey depth, critical analysis and synthesis, and oral presentation clarity with slide design.

8. Institutional Letter Grading Scale & SGPA/CGPA Computation

Northgate Institute adopts a 10-point absolute grading system with relative moderation when statistically justified. Conversion: 90–100 $\rightarrow$ O (Outstanding, 10.0); 80–89 $\rightarrow$ A+ (9.0); 70–79 $\rightarrow$ A (8.0); 60–69 $\rightarrow$ B+ (7.0); 55–59 $\rightarrow$ B (6.0); 50–54 $\rightarrow$ C (5.0); $< 50 \rightarrow$ F (0.0). $SGPA = \sum(Credit_i \times GradePoint_i) / \sum Credit_i$ for the current semester. $CGPA = \sum(Credit_i \times GradePoint_i) / \sum Credit_i$ across all completed semesters. Repeated course grades replace the prior F grade in the CGPA computation.

9. Grade Moderation, Statistical Parity & Inter-Section Equity

The Departmental Grade Moderation Committee (GMC) convenes within 10 days of end-semester result preparation. The GMC performs: (a) cross-section comparison of mean, median, and standard deviation for each course; (b) identification of statistically significant inter-section disparities ($p < 0.05$ on Mann-Whitney U test); (c) application of linear scaling adjustments or addition of a uniform scaling constant when inter-section parity violations are confirmed; (d) mandatory review of any course with a failure rate exceeding 30% to determine whether course design, assessment calibration, or instructional factors require intervention.

10. Supplementary Examinations & Grade Improvement Policy

Students earning an 'F' grade may appear for the Supplementary Examination (conducted during the subsequent term break). The maximum grade achievable through a supplementary exam is 'B+' (7.0 grade points), regardless of marks scored, to maintain fairness with students who cleared the course in the regular examination. Students wishing to improve a passing grade (C or B) may re-register for the course in the next available offering, forfeiting the prior grade. The improved grade replaces the original grade in the CGPA computation from the next semester results onwards.

11. AI Proctoring, Remote Assessment & Digital Examination Integrity

For online assessments conducted via the DCMP LMS, the University employs AI-based proctoring tools that monitor: (a) face detection and identity verification via webcam; (b) screen recording with keystroke logging for suspicious activity (rapid copy-paste, browser tab switching); (c) audio monitoring for voice input or communication with third parties. AI proctoring flags are reviewed by the DAIC before any disciplinary action is initiated. False positives must be appealed within 48 hours of the assessment conclusion.